

AYMANE AIT DADS

Data Science Intern - Simulation Acceleration | Geometric Deep Learning · PyTorch · Physics-Based Modeling
Aymane.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Engineering student with hands-on PyTorch deep learning experience, including training a 30B Mamba-Transformer MoE model and building multi-architecture CV pipelines on 250K-image datasets, directly applicable to ML-based CFD simulation acceleration. Reduced manual analysis time by 60% at Orange Maroc by engineering automated ML pipelines on 100K+ network records, demonstrating structured experiment design and measurable outcomes. Proficient in Python, PyTorch, deep learning frameworks, ablation studies, and data preprocessing workflows aligned with the Fast Physics project requirements. Fully available for a full-time summer internship in Europe; no work authorization restrictions apply.

CORE COMPETENCIES

Geometric Deep Learning | Physics-Based Modeling | PyTorch Deep Learning | CFD Simulation Data | Graph Neural Networks | Experiment Validation | Python Data Preprocessing | Simulation Acceleration

WORK EXPERIENCE

Orange Maroc Jul 2024 - Aug 2024
Data Science Intern - Casablanca, Morocco

- Engineered an automated Python ML pipeline combining Random Forest and K-Means on **100K+ fiber-optic network records**, classifying quality across thousands of nodes with zero manual intervention.
- Developed a clustering framework mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis and reporting time by **~60%** through automated feature engineering, model evaluation scripts, and structured experiment documentation in Python and Scikit-learn.
- Delivered stakeholder presentations translating model outputs into **actionable maintenance recommendations**, communicating technical findings to non-technical business teams via Power BI dashboards.

PROJECTS

Anomalous Sound Detection - Industrial Equipment EURECOM · 2025 · AUC 0.80 Predictive Maintenance

- Designed an unsupervised Transformer autoencoder on Mel spectrograms with SpecAugment for slide-rail machinery fault detection, achieving **AUC 0.80** with no labeled anomaly data available.
- Preprocessed raw industrial sensor signal data into structured numerical representations, ran systematic validation experiments in PyTorch, and documented findings aligned with **physics-based modeling** objectives.

PyTorch · Transformer Autoencoder · SpecAugment · Mel Spectrograms

AI-Generated Image Detection - NTIRE 2026 @ CVPR EURECOM + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st EURECOM Class

- Ranked 1st on the EURECOM class private Kaggle leaderboard (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head on **250K images from 25+ generator types**, then submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Conducted structured ablation studies isolating epoch count, unfreezing depth, and ensemble weighting, confirming that **1-epoch training outperforms 4-epoch** by 2.6 AUC points for out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle T4

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM Sep 2023 - Present
Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Statistics, Cloud Computing, NLP, Image Security

CPGE - Mathematics & Physics - Ibn Timiya 2021 - 2023

SKILLS

ML & Deep Learning (Physics / Geometry Focus): PyTorch | Scikit-learn | TensorFlow | Hugging Face Transformers | CNN | ViT | DenseNet | ablation studies

Data Science & Preprocessing: Python | Pandas | NumPy | feature engineering | clustering | ensemble methods | SQL | Matplotlib

MLOps & Infrastructure: Git | Linux | FP16/BF16 mixed precision | CUDA debugging | Docker (basic) | Kaggle GPU | Power BI